DATA STRUCTURES HOMEWORK 1

1. Sum of Two Matrices

CODE

```
#include <stdio.h>
int main(){
    int mat1[10][10], mat2[10][10], rows, cols, i, j;
    printf("Enter size of row:");
    scanf("%d",&rows);
    printf("Enter size of column:");
    scanf("%d",&cols);
    printf("Enter elements in matrix 1:\n");
    for(i=0; i<rows; i++){
        for(j=0; j<cols; j++){
            scanf("%d",&mat1[i][j]);
        }
    }
    printf("Enter elements in matrix 2:\n");
    for(i=0; i<rows; i++){
        for(j=0; j<cols; j++){
            scanf("%d",&mat2[i][j]);
        }
    }
    printf("\n");
    printf("Matrix 1:\n");
    for(i=0; i<rows; i++){
        for(j=0; j<cols; j++){
            printf("%d ",mat1[i][j]);
        }
        printf("\n");
    }
    printf("\n");
    printf("Matrix 2:\n");
    for(i=0; i<rows; i++){
        for(j=0; j<cols; j++){
            printf("%d ",mat2[i][j]);
        }
        printf("\n");
    }
    printf("\n");
    printf("Sum of two matrices:\n");
    for(i=0; i<rows; i++){
        for(j=0; j<cols; j++){
            printf("%d ",mat1[i][j] + mat2[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
Enter size of row:2
Enter size of column:2
Enter elements in matrix 1:
1
2
3
4
Enter elements in matrix 2:
5
6
7
8
Matrix 1:
1 2
3 4
Matrix 2:
5 6
7 8
Sum of two matrices:
6 8
10 12
```

2. Read n Values and Display in Reverse Order

CODE

```
#include <stdio.h>
int main(){
    int arr[100];
    int n,i;
    printf("Enter the size of array:");
    scanf("%d",&n);
    printf("Enter %d elements: \n",n);
    for(i=0; i<n; i++){
        scanf("%d",&arr[i]);
    }
    printf("Array elements in Reverse order:\n");
    for(i=n-1; i>=0; i--){
        printf("%d ",arr[i]);
    }
    return 0;
}
```

OUTPUT

```
Enter the size of array:5
Enter 5 elements:
12
34
78
65
98
Array elements in Reverse order:
98 65 78 34 12
```

3. Count Total Number of Duplicate Elements

CODE

```
#include <stdio.h>
int main(){
    int arr[100], n, i, j, dup_count = 0;
    printf("Enter the size of array:");
    scanf("%d",&n);
    printf("Enter %d elements: \n",n);
    for(i=0; i<n; i++){
        scanf("%d",&arr[i]);
    }
    for(i = 0; i < n; i++) {
        for(j = 0; j < i; j++) {
            if(arr[i] == arr[j]) {
                dup_count++;
                break;
            }
        }
    }
    printf("Total duplicate elements in array: %d\n",dup_count);
    return 0;
}
```

OUTPUT

```
Enter the size of array:5
Enter 5 elements:
11
12
11
43
12
Total duplicate elements in array: 2
```

4. Print Unique Elements in an Array

CODE

```
#include <stdio.h>
int main(){
    int arr[100], n, i, j, count;
    printf("Enter the size of array:");
    scanf("%d",&n);
    printf("Enter %d elements: \n",n);
    for(i=0; i<n; i++){
        scanf("%d",&arr[i]);
    }
    printf("Unique elements in the array: ");
    for(i=0; i<n; i++){
        count=0;
        for(j=0; j<n; j++){
            if(arr[j] == arr[i]){
                count++;
            }
        }
        if(count == 1) {
            printf("%d ", arr[i]);
        }
    }
    return 0;
}
```

OUTPUT

```
Enter the size of array:8
Enter 8 elements:
12
34
56
12
11
12
10
68
Unique elements in the array: 34 56 11 10 68
```

5. Insert in Sorted Array

CODE

```
#include <stdio.h>
int main(){
    int arr[100],n,i,val;
    printf("Enter the size of sorted array:");
    scanf("%d",&n);
    printf("Enter %d elements: \n",n);
    for(i=0; i<n; i++){
        scanf("%d",&arr[i]);
    }
    printf("Enter the value:");
    scanf("%d",&val);
    for(i=n-1; i>=0 && arr[i]>val; i--){
        arr[i+1]=arr[i];
    }
    arr[i+1] = val;
    n++;
    printf("New array: \n");
    for(i=0; i<n; i++){
        printf("%d ",arr[i]);
    }
    return 0;
}
```

OUTPUT

```
Enter the size of sorted array:5
Enter 5 elements:
12
45
56
67
89
Enter the value:34
New array:
12 34 45 56 67 89
```

6. Insert at a Given Position in an Array

CODE

```
#include <stdio.h>
int main() {
    int arr[100], n, i, pos, value;
    printf("Enter the size of array:");
    scanf("%d",&n);
    printf("Enter the elements:");
    for (i=0; i<n; i++){
        scanf("%d",&arr[i]);
    }
    printf("Enter the position to insert:");
    scanf("%d",&pos);
    printf("Enter the value:");
    scanf("%d",&value);
    i = n - 1;
    while(i >= pos-1){
        arr[i+1] = arr[i];
        i--;
    }
    arr[pos-1] = value;
    n = n+1;
    printf("New array:\n");
    for (i=0; i<n; i++){
        printf("%d ",arr[i]);
    }
    return 0;
}
```

OUTPUT

```
Enter the size of array:5
Enter the elements:
12
34
56
78
90
Enter the position to insert:3
Enter the value:64
New array:
12 34 64 56 78 90
```

7. Matrix Multiplication

CODE

```
#include <stdio.h>
int main() {
    int r1, c1, r2, c2, i, j, k;
    printf("Enter rows of first matrix: ");
    scanf("%d", &r1);
    printf("Enter columns of first matrix: ");
    scanf("%d", &c1);
    printf("Enter rows second matrix: ");
    scanf("%d", &r2);
    printf("Enter columns of first matrix: ");
    scanf("%d", &c2);

    if (c1 != r2) {
        printf("Matrix multiplication is not possible.\n");
        return 0;
    }

    int A[r1][c1], B[r2][c2], C[r1][c2];

    printf("Enter elements of first matrix:\n");
    for (i = 0; i < r1; i++)
        for (j = 0; j < c1; j++)
            scanf("%d", &A[i][j]);

    printf("Enter elements of second matrix:\n");
    for (i = 0; i < r2; i++)
        for (j = 0; j < c2; j++)
            scanf("%d", &B[i][j]);

    for (i = 0; i < r1; i++)
        for (j = 0; j < c2; j++)
            C[i][j] = 0;

    for (i = 0; i < r1; i++) {
        for (j = 0; j < c2; j++) {
            for (k = 0; k < c1; k++) {
                C[i][j] += A[i][k] * B[k][j];
            }
        }
    }

    printf("Resultant Matrix:\n");
    for (i = 0; i < r1; i++) {
        for (j = 0; j < c2; j++) {
            printf("%d ", C[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
Enter rows of first matrix: 2
Enter columns of first matrix: 2
Enter rows second matrix: 2
Enter columns of first matrix: 2
Enter elements of first matrix:
1
2
3
4
Enter elements of second matrix:
5
6
7
8
Resultant Matrix:
19 22
43 50
```

8. Matrix Transpose

CODE

```
#include <stdio.h>
int main(){
    int arr[10][10], rows, cols, i, j;
    printf("Enter size of row:");
    scanf("%d",&rows);
    printf("Enter size of column:");
    scanf("%d",&cols);
    printf("Enter elements of the matrix:\n");
    for(i=0; i<rows; i++){
        for(j=0; j<cols; j++){
            scanf("%d",&arr[i][j]);
        }
    }
    printf("\nMatrix:\n");
    for(i=0; i<rows; i++){
        for(j=0; j<cols; j++){
            printf("%d ",arr[i][j]);
        }
        printf("\n");
    }
    printf("Transpose of the matrix:\n");
    for(i=0; i<cols; i++){
        for(j=0; j<rows; j++){
            printf(" %d ",arr[j][i]);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
Enter size of row:2
Enter size of column:3
Enter elements of the matrix:
12
34
45
56
67
78
Matrix:
12 34 45
56 67 78
Transpose of the matrix:
12 56
34 67
45 78
```

9. Right Diagonal Sum

CODE

```
#include <stdio.h>
int main(){
    int arr[10][10], n, i, j, sum=0;
    printf("Enter size of matrix:");
    scanf("%d",&n);
    printf("Enter elements of the matrix:\n");
    for(i=0; i<n; i++){
        for(j=0; j<n; j++){
            scanf("%d",&arr[i][j]);
        }
    }
    printf("\nMatrix:\n");
    for(i=0; i<n; i++){
        for(j=0; j<n; j++){
            printf("%d ",arr[i][j]);
        }
        printf("\n");
    }
    for(i = 0; i < n; i++) {
        for(j = 0; j < n; j++) {
            if(i + j == n - 1) {
                sum += arr[i][j];
            }
        }
    }
    printf("\nRight diagonal sum = %d", sum);
    return 0;
}
```

OUTPUT

```
Enter size of matrix:3
Enter elements of the matrix:
10
11
12
23
45
76
34
14
15
Matrix:
10 11 12
23 45 76
34 14 15
Right diagonal sum = 91
```

10. Check for Identity Matrix

CODE

```
#include <stdio.h>
int main(){
    int arr[10][10], n, i, j, found=1;
    printf("Enter size of matrix:");
    scanf("%d",&n);
    printf("Enter elements of the matrix:\n");
    for(i=0; i<n; i++){
        for(j=0; j<n; j++){
            scanf("%d",&arr[i][j]);
        }
    }
    printf("\nMatrix:\n");
    for(i=0; i<n; i++){
        for(j=0; j<n; j++){
            printf("%d ",arr[i][j]);
        }
        printf("\n");
    }
    for(i=0; i<n; i++){
        for(j=0; j<n; j++){
            if(i == j){
                if(arr[i][j] != 1)
                    found = 0;
            }
            else{
                if(arr[i][j] != 0)
                    found = 0;
            }
        }
    }
    if(found==1)
        printf("Identity matrix");
    else
        printf("Not an identity matrix");
    return 0;
}
```

OUTPUT

```
Enter size of matrix:3
Enter elements of the matrix:
1
0
0
0
1
0
0
0
1
Matrix:
1 0 0
0 1 0
0 0 1
Identity matrix
```

11. Count Vowels, Consonants, Digits and White Spaces

CODE

```
#include <stdio.h>
int main(){
    char str[100];
    int i;
    int digits=0, vowels=0, con=0, spaces=0;

    printf("Enter the string: ");
    fgets(str, sizeof(str), stdin);

    for(i=0; str[i] != '\0'; i++){
        if(str[i] >= '0' && str[i] <= '9'){
            digits++;
        }
        else if(str[i] == 'a' || str[i] == 'e' || str[i] == 'i' ||
                str[i] == 'o' || str[i] == 'u' || str[i] == 'A' ||
                str[i] == 'E' || str[i] == 'I' || str[i] == 'O' ||
                str[i] == 'U'){
            vowels++;
        }
        else if(str[i] == ' '){
            spaces++;
        }
        else if((str[i] >= 'A' && str[i] <= 'Z') ||
                (str[i] >= 'a' && str[i] <= 'z')){
            con++;
        }
    }

    printf("Digits: %d\nVowels: %d\nConsonants: %d\nWhite spaces: %d",
           digits, vowels, con, spaces);
    return 0;
}
```

OUTPUT

```
Enter the string: I am very much interested in Problem solving 5028
Digits: 4
Vowels: 13
Consonants: 24
White spaces: 8
```

12. Frequency of Character in a String

CODE

```
#include <stdio.h>
int main(){
    char str[100];
    int i, freq[26] = {0};

    printf("Enter the string: ");
    fgets(str, sizeof(str), stdin);

    for(i=0; str[i] != '\0'; i++){
        if(str[i] >= 'A' && str[i] <= 'Z'){
            str[i] = str[i] + 32;
        }
    }

    for(i=0; str[i] != '\0'; i++){
        if(str[i] >= 'a' && str[i] <= 'z'){
            freq[str[i] - 'a']++;
        }
    }

    for(i=0; i<26; i++){
        if(freq[i] > 0){
            printf("%c = %d\n", i + 'a', freq[i]);
        }
    }
    return 0;
}
```

OUTPUT

```
Enter the string: programming
a = 1
g = 2
i = 1
m = 2
n = 1
o = 1
p = 1
r = 2
```

13. Count Words in a String

CODE

```
#include <stdio.h>
int main(){
    char sen[100];
    int i, words=1;

    printf("Enter the sentence:");
    fgets(sen, sizeof(sen), stdin);

    if(sen[0] == '\0' || sen[0] == '\n'){
        words = 0;
    }

    for(i=0; sen[i] != '\0'; i++){
        if(sen[i] == ' ' && sen[i+1] != ' ' &&
           sen[i+1] != '\0'){
            words++;
        }
    }

    printf("Number of words in the given sentence:%d",words);
    return 0;
}
```

OUTPUT

```
Enter the sentence:I am very much interested in Problem solving
Number of words in the given sentence:8
```

14. Sort String Array

CODE

```
#include <stdio.h>
#include <string.h>
int main() {
    char str[10][50], temp[50];
    int n, i, j;

    printf("Enter the number of strings: ");
    scanf("%d", &n);
    getchar();

    printf("Enter the strings:\n");
    for(i = 0; i < n; i++) {
        fgets(str[i], sizeof(str[i]), stdin);
        str[i][strcspn(str[i], "\n")] = '\0';
    }

    for(i = 0; i < n - 1; i++) {
        for(j = i + 1; j < n; j++) {
            if(strcmp(str[i], str[j]) > 0) {
                strcpy(temp, str[i]);
                strcpy(str[i], str[j]);
                strcpy(str[j], temp);
            }
        }
    }

    printf("\nSorted strings:\n");
    for(i = 0; i < n; i++) {
        printf("%s\n", str[i]);
    }
    return 0;
}
```

OUTPUT

```
Enter the number of strings: 3
Enter the strings:
Mango
Grapes
Orange
Sorted strings:
Grapes
Mango
Orange
```

15. Toggle Case of a Sentence

CODE

```
#include <stdio.h>
int main(){
    char str[100];
    printf("Enter string:");
    fgets(str, sizeof(str), stdin);

    for(int i = 0; str[i] != '\0'; i++){
        if(str[i] >= 'A' && str[i] <= 'Z'){
            str[i] = str[i] + 32;
        }
        else if(str[i] >= 'a' && str[i] <= 'z'){
            str[i] = str[i] - 32;
        }
    }

    printf("Case changed string:%s",str);
    return 0;
}
```

OUTPUT

```
Enter string:HELlo woRLd
Case changed string:helLO WOrlD
```